FLOODGATE

Team Name: Bitheads
Problem Statement: Enabling strategic foresight for investors and small business owners through AI-powered forecasting and advisory systems.

Project Overview

FloodGate is an intelligent forecasting platform designed to help investors and small business owners anticipate future risks and opportunities. It combines time-series prediction models with large language models (LLMs) to deliver actionable insights and strategic guidance. The platform features a user-friendly web interface that presents forecasts and recommendations in a clear, interactive format.

Frontend Design Concept

The frontend follows a modern, minimalistic, and data-centric design approach. The interface includes sidebar navigation, forecast cards, a search function, and a dynamic advisory panel. AI tools such as the OpenAI API, Copilot, FastAPI, Gemini, and ChatGPT were used to generate color palettes, layout prototypes, and CSS snippets. Human input refined these suggestions through improved spacing, accessibility adjustments, custom hover effects, and enhanced readability for a polished user experience.

Backend Planning

The backend is built using **FastAPI**, with **CSV-based local storage** that can scale to Firebase or PostgreSQL. The key AI integration point is the **/advice** endpoint, which connects to the OpenAI API for generating recommendations. Core backend functions include forecast generation, AI advisory logic, and RESTful API endpoints. ChatGPT supported logic scaffolding, while GitHub Copilot accelerated code completion and integration.

AI Pipeline and Model Integration

FloodGate integrates AI models such as **OpenAI GPT-4** and **Prophet** for predictive analytics and strategic advisory generation. Data flows from the user through the frontend to the backend, then to the AI models, and finally returns as structured insights to the frontend. AI handles prediction and contextual recommendation tasks, while humans fine-tune prompts, validate outputs, and adjust logic to ensure relevance and clarity.

API and Data Pipeline

The system integrates both internal (FastAPI) and external (OpenAI) APIs. Data primarily uses CSV formats. The processing pipeline moves from the frontend to the backend, through the AI layer, and back to the frontend. Custom validation logic handles errors and fallback messages, while human oversight ensures accuracy in edge cases.

AI Tools and Their Roles

ChatGPT ,Gemini was used to generate backend logic and UI scaffolding, providing forecast and advisory endpoints optimized for clarity and performance.
GitHub Copilot assisted with frontend and backend code completion, producing function snippets and boilerplate that were then manually refined and tested.

Integration Architecture

The overall system flow follows a simple yet robust sequence:
User → Frontend (React) → Backend (FastAPI) → AI API (OpenAI/Gemini) → Backend → UI Output.
This architecture ensures seamless data transfer and real-time response generation.

Human Touch and Creative Input

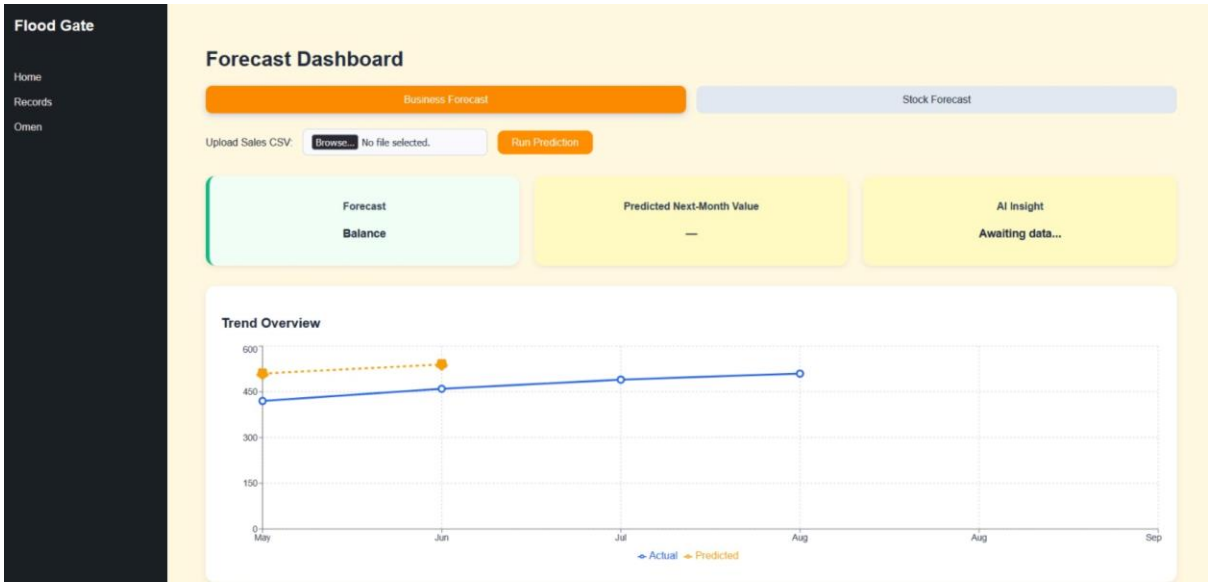
Human creativity played a key role throughout development. The team manually refined navigation flow, enhanced interactivity with custom hover effects and card animations, and tuned AI prompts for domain-specific clarity. The color palette was adjusted for accessibility, and the user experience was designed to intuitively support both investors and small business owners.

Implementation Details

The frontend was developed using **HTML, CSS, and JavaScript** and react. Full integration between frontend and backend is achieved through FAST API endpoints such as **/forecast** and **/advice**.

Innovation Highlights

FloodGate introduces a dual-persona forecasting engine customized for both investors and small businesses. It employs an agentic AI architecture that merges predictive modeling with LLM-based strategic guidance.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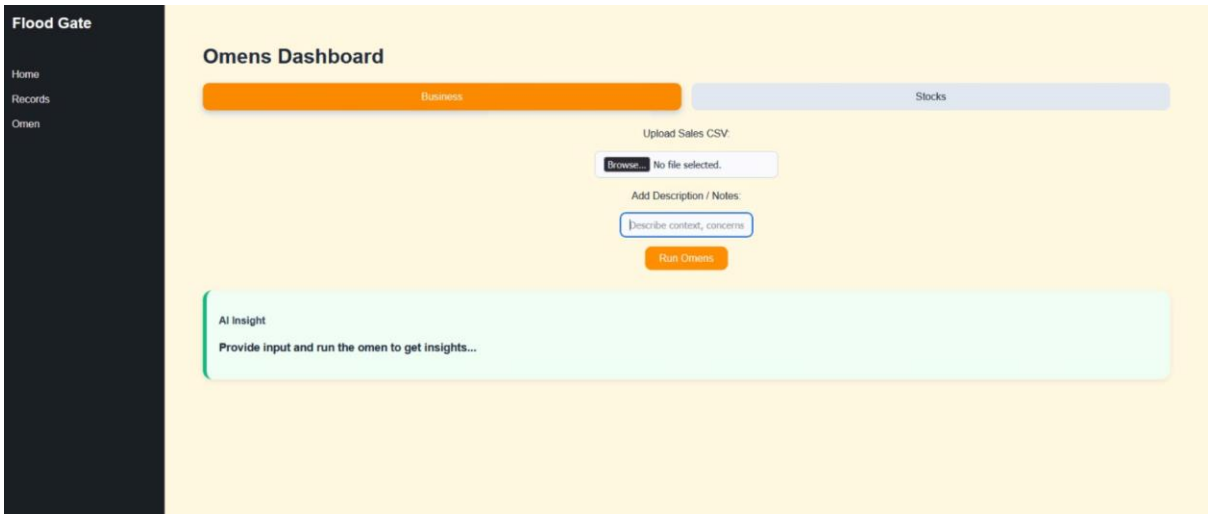
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Upload Sales CSV:
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